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Symmetric Measure of Network Traffic using Packet Ratio and Packet Symmetry

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Abstract

Flood attacks on a network occur when attackers send a very high volume of traffic to a system. This leads to an exhaustion of resources of the targeted system. Such a system or infrastructure under attack won't be able to cope up with the services it needs to provide due to capacity overload. These types of attacks are also called as Denial of Service (DoS) attacks. As it is essential for companies and organizations to provide high speed and high-quality services the network in which it operates needs to be secured against DoS attacks. As it is said prevention is better than cure, it is always better to detect an attack in the preliminary stages itself and deny an attacker any opportunity to continue the assault on the system. There are various kinds of DoS attacks, in this study the focus is on three kinds of attacks and its detection ICMP flood attack, UDP flood attack and TCP SYN flood attack. The detection mechanism employed is based on packet symmetry which can be used in to monitor the data flow in a network and detect any malicious anomalies in the network traffic at any instant. The experimental results, show that the packet symmetry is a reliable metric to detect attacks on the network by checking the data flow in the ICMP, UDP and TCP layers.

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Keywords: Denial of Service; packet symmetry; transmission ratio;ICMP;UDP;TCP-SYN

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1. Introduction

1.1. Need for providing high quality network service

Internet has been growing for the past decade at a high rate. Its availability around the world has also growing rapidly. As a matter of fact it has become a part of our lives nowadays. Most of the business deals and events take place through the internet. It is hard for people to accept a slow, low quality internet service. Due to the high speed transactional nature of today's networking it is essential to run any network at its full potential.

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1.3. Flood attacks

Flooding attacks or DoS attacks have always been a threat to networks [1] In 2016 a DDoS attack against Dyn, a major DNS provider was very devastating and created disruption for many major sites like Netflix, PayPal, Reddit, Visa, AirBnB, GitHub etc. Such attacks can disrupt the normal flow of network resulting in slower services. This in turn can cause loss of reputation for the affected company as well as large amount of time and money [2]. In comparison to hacking attacks, DoS attacks usually doesn't try to create a security breach or steal information. Due to this nature of DoS attacks it is essential to detect such attacks at the earliest to prevent these attacks before it costs the service provider a great fortune [3]. Most of the companies to detect such attacks use Network Intrusion Detection Systems (NIDS).

1.4. Focus of study

In this study, three different kinds of attacks which are ICMP, UDP and TCP-SYN attacks are focussed. Our focus was to use packet symmetry to find anomalies in pcap files. The metric was applied to normal and attack files of ICMP, UDP and TCP packet layers and the comparison between the normal and attack files have been studied. The main attempt was to see how the packet ratio varied during an attack and whether it can be used to detect an attack. Also, the difference between packet transmissions and packet symmetry across time has been studied. The work using packet symmetry tries to differentiate a good network traffic as a symmetrical one comparing the packets send and received – from source to destination and vice versa. Such a network has a balance of packets across both sides whereas a bad network traffic which in turn corresponds to asymmetry due to imbalance across endpoints. Especially in UDP layer of network there are no effective method to check for packet requests and replies due to a lack of any kind of handshake mechanisms as in TCP layer, the use of packet symmetry can be a helpful method to identify anomalies in the network.

2. Related work

The connection and flow statistics are essential to detect the flooding attacks. It detects all kinds of DDoS attack such as TCP, UDP and ICMP flood attacks. Further, rate limiting is applied to mitigate the flows that are aggressive, and quality of service is improved [3]. Packet symmetry and packet ratio provides efficient results in detecting the DDoS attacks for various kinds such as TCP, UDP and ICMP [4] [5]. The attacks that are intended to consume the server resources by sending aggressive traffic flows are observed by the metrics and depicts the presence of DDoS attacks. Backscatter analysis is used for determining the DoS attack in the various domains. DDoS attacks is distributed and originates from various ISPs and this wide activity is studied using the backscatter

technique by using various protocol exploit attacks such as HTTP, TCP, UDP and ICMP. The attack size and length of the denial-of-service attack is analyzed, and it is evident that attack duration is long and distributed flows constitutes heavy traffic at the ISP networks [6]. DDoS attack is detected using various passive measurements that identifies the asymmetry in the protocol exploit attacks. Number of flows that are active, rate at which the flow is generated, and policies are used to identify the spoofed attacks. Also, the size and length of the flow distribution clearly indicates the presence of spoofed attacks in the network [7]. Flow and connection statistics is used to identify the DDoS attack and provide information security [8].

3. Proposed system

The proposed system includes modules such as packet processing, packet symmetry computation and packet ratio computation. The processing system uses the data that are free from redundancy and only the major attributes are passed to the computation modules. The computation modules use packet symmetry and packet ratio to identify the malicious traffic in the network. The visualization results provide a clear picture of the flow statistics of TCP flood, ICMP flood and UDP flood attacks.

3.1. Transmission per second

The Network Traffic refers to the flow of data moving across a network at a given time. The same can also be known as packet transmission over the network per second. With the targeted host being flooded with malicious / non-malicious packets, a possible metric to detect a possible flood attack, it to check the transmissions taking place per second, and therefore set a probable threshold to set an alarm. In Wireshark, this metric can be graphed under Conversations, i.e. the traffic between two IP addresses. The number of packets conversed in a t second window ($t = 1$) and under what protocol the packet was sent is being calculated. The protocol could be UDP, ICMP, and TCP. This metric though viable is not a good metric because, the threshold being set is static, but the network traffic is dynamic. Therefore, it is biased and will not work well.

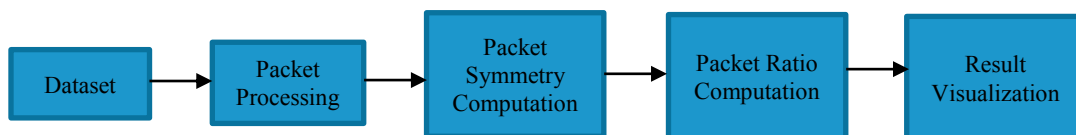


Fig. 1. Workflow of the proposed system.

3.2. Transmission per second packet symmetry / packet asymmetry

This new metric allows an unbiased evaluation of the traffic centered around. The UDP protocol is used in applications such as gaming, video streams, Voice over IP (VoIP), where little delays between sent and received packets are necessary [9]. Packet transmission metric is the most important metric, because it serves as the indicator that DDoS attack is occurring in the cloud. The ratio between the packets sent and received clearly indicates the abnormal rise in the traffic behaviour. The significance of this metric is validated by analysing both the normal and attack files. The baseline profile can be understood with the normal files and any deviations are recorded as the attack. Even during the huge work load the valid requests are increasing with the time, packet ratio can clearly depict the presence of flooding attacks. Packet Symmetry is a method that bases the symmetry of the network [10] connection on the number of sent and received packets as in Equation 1.

$$S = \log \left(\frac{tx+1}{rx+1} \right) \quad (1)$$

S –Packet Symmetry Metric which is centered around 0

tx - number of sent packets and

rx is number of received packets per unit time.

When S is fluctuating from 0, then the network flow could be argued as malicious, because for every packets received there should be a packet sent as an acknowledgement, therefore during a normal network connection the balance between the sent and received is maintained as in Figure 2.

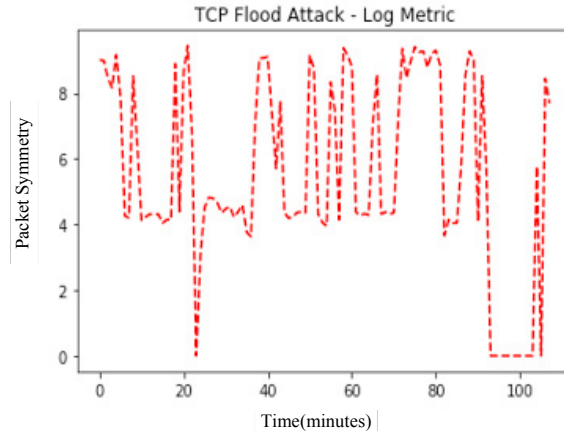


Fig. 2. Packet symmetry (s) on tcp flood attack packet capture file.

3.3. Packet ratio

The packet ratio is quite similar to packet symmetry, where the metric is showing the asymmetry between the numbers of transmitted and received packets as in Equation 2. Here if the, packets in the traffic are balanced, then the ratio will be one, therefore this metric is centered around 1 as in Figure 3.

$$R = \begin{cases} \frac{tx}{rx}, & \text{if } tx > 0 \\ M, & \text{if } tx = 0 \end{cases} \quad (2)$$

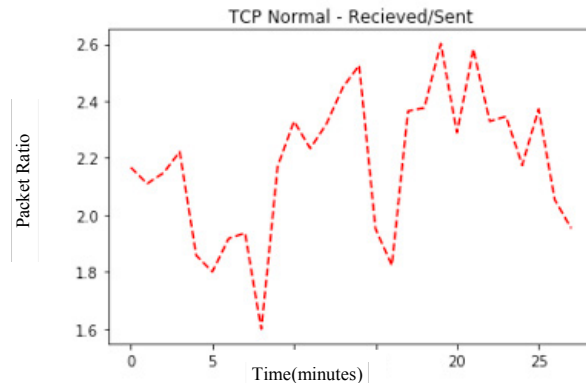


Fig. 3. Packet ratio in tcp-syn normal packet capture file

4. Results and discussions

The openstack cloud is configured with three attacker nodes and one target nodes. The flooding packets of various protocols such as TCP, UDP and ICMP is launched to the target virtual instance, where the traffic logs are collected in real time to study the attack characteristics. Since, the real time data is utilized for understanding the metrics such as packet symmetry and packet ratio, the asymmetrical behavior of the DDoS attack traffic can be investigated. The cloud traces are captured with the pcap files of size approximately 1 GB. These files are then processed by the proposed system to indicate the DDoS traffic in cloud environment.

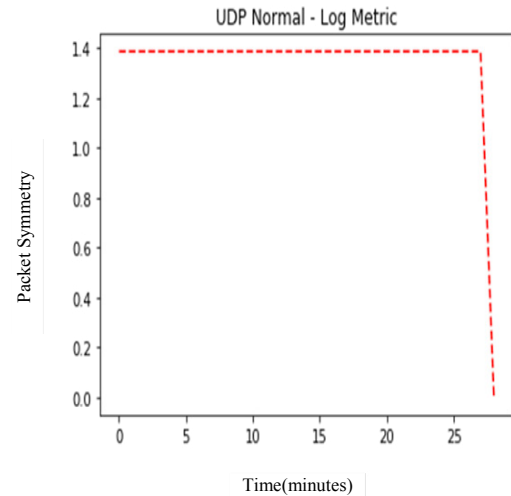
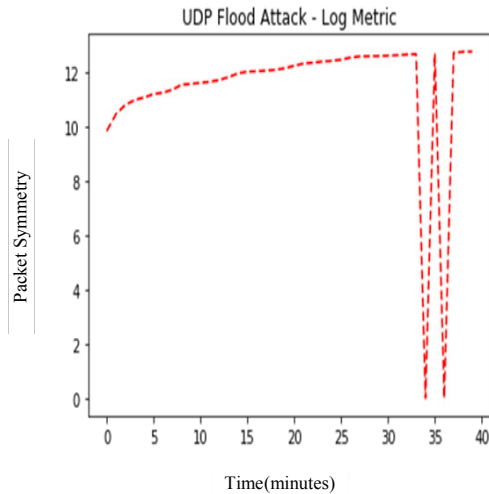


Fig. 4. Packet Symmetry on UDP Flood Attack.pcap Fig. 5. Packet Symmetry on UDP Normal.pcap

The following metrics are allowing us to detect the possibility of an attack based on just the data traffic flow. The packet symmetry metrics on UDP flood and ICMP attack depicts the flow characteristics have abnormalities. The abnormal peaks and drops in the statistics clearly state the flows are not symmetrical as shown in Figure 4, Figure 6, Figure 7 and Figure 9. The packet symmetry on UDP and ICMP flow metric shows that the traffic is normal, and the flows are exhibiting the normal activity as shown in Figure 5 and Figure 8. The proposed system confirms that it is well suited to detect UDP flood attacks, so the testing is extended to TCP and ICMP protocols and found that the metrics are able to show a dissimilarity in the observed traffic.

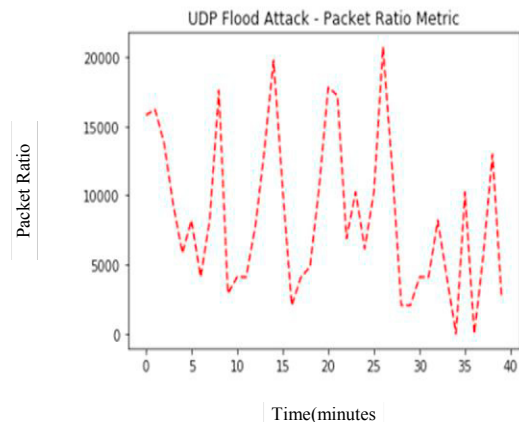
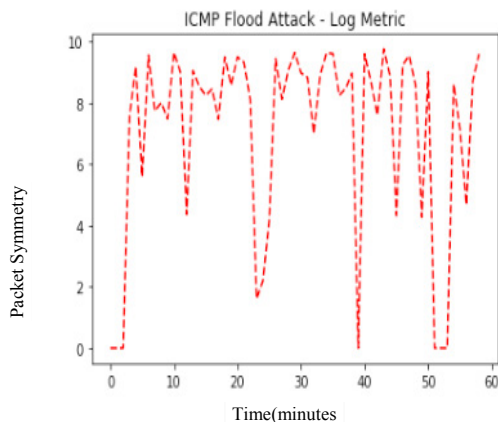


Fig. 6. Packet Symmetry on ICMP Flood.pcap

Fig. 7. Packet Ratio on UDP Flood Attack.pcap

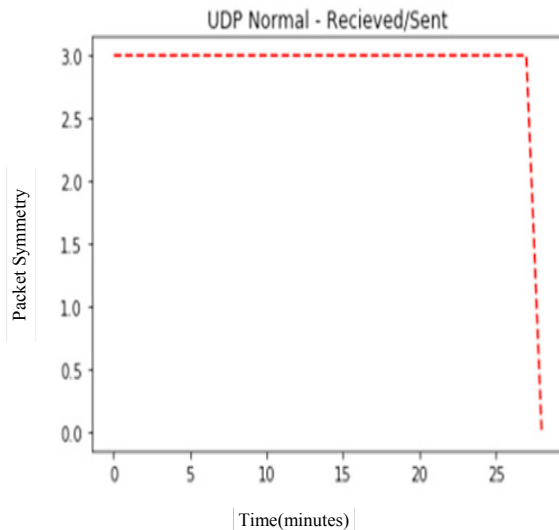


Fig. 8. Packet ratio on UDP Normal.pcap

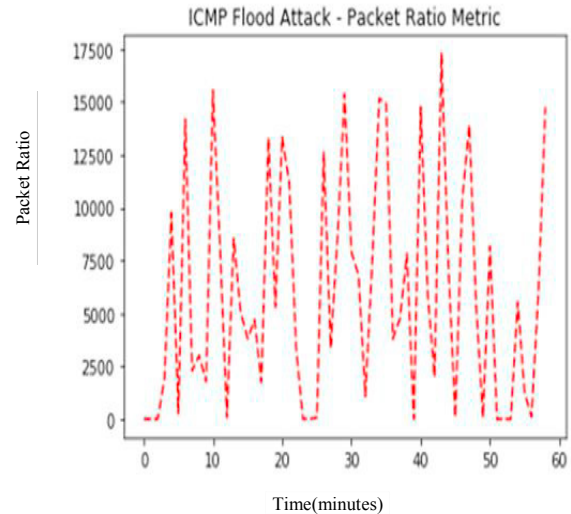


Fig. 9. Packet ratio on ICMP Flood Attack.pcap

The packet computation techniques are used to determine the packet symmetry/packet ratio for traffic monitoring in the cloud and it detects any DDoS attack that interrupts the cloud availability. The proposed system utilizes the packet symmetry and packet ratio as the major parameter to detect the malicious traffic. The variants of attacks studied are ICMP flood, UDP flood and TCP SYN flood. From the experimental results, the passive measurements are vital to distinguish the DDoS attack from the normal in the cloud computing environment. Future work is to apply the measurements for the larger dataset and involve learning mechanism to differentiate the benign applications and malicious.

5. Conclusion and future work

The proposed system uses the packet symmetry and packet ratio passive measurement heuristics as means of malicious traffic detection and aims to show the usefulness of such an approach. The various kinds of DDoS attacks, used in this study are ICMP flood attack, UDP flood attack and TCP SYN flood attack. The packet computation methods are employed based on packet symmetry/packet ratio for monitoring flow in the network and it detects any malicious traffic that affects the availability of cloud services. Hence, the passive measurements are very essential to detect the DDoS attack by using the reliable metrics. Future work is to apply the measurements for the larger dataset and involve learning mechanism to differentiate the benign applications and malicious.

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